Stochastic Processes Semester 1

Tutorial 2 – Solutions

1. (a) If $S_{10} \geq 0$ then $X_k = 1$ at least 5 times for $k \leq 10$.
   So we have

$$\begin{aligned}\mathbf{P}(S_{10} \geq 0) &= \sum_{k=5}^{10} \binom{10}{k} \left(\frac{2}{3}\right)^k \left(\frac{1}{3}\right)^{10-k} \\ &\approx 0.923.\end{aligned}$$

   (b) We have $\mathbf{E}(S_{100}) = 100\mathbf{E}(X_k) = 100/3$ and $\mathbf{Var}(S_{100}) = 100\mathbf{Var}(X_k) = 800/9$. Hence, using the normal approximation to the binomial distribution we have

$$\begin{aligned}\mathbf{P}(S_{100} \geq 30) &= \mathbf{P}\left(\frac{S_{100} - 100/3}{\sqrt{800/9}} \geq \frac{30 - 100/3}{\sqrt{800/9}}\right) \\ &\approx 1 - \Phi(-0.3536) = 0.638.\end{aligned}$$

   (Here, as usual, Φ is the (cumulative) distribution function of the $N(0, 1)$ distribution.)

   (c) The required probability is the same as the probability that the same random walk, but starting at $S_0 = 2$, ever hits state 0. This is just the probability of ruin for the latter process and so is equal to $(q/p)^2$, which here is $1/4$.

2. The fortune of the insurance company is a random walk $S_n = u + \sum_{i=1}^{n} X_i$, where $X_1, X_2 \ldots$ are independent identically distributed random variables with

$$\begin{aligned}\mathbf{P}(X_i = 1) &= 1/2 \\ \mathbf{P}(X_i = 0) &= 1/4 \\ \mathbf{P}(X_i = -1) &= 1/4.\end{aligned}$$

   Hence the probability that the company ever runs out of capital is

$$\min\left(1, \left(\frac{1/4}{1/2}\right)^u\right) = \left(\frac{1}{2}\right)^u = 0.0078$$

   Hence, $u = 7$.

3. (a) The fortune of the insurance company is a random walk $S_n = u + \sum_{i=1}^{n} X_i$, where $X_1, X_2 \ldots$ are independent identically distributed random variables with

$$\begin{aligned}\mathbf{P}(X_i = 2) &= 1/20\\ \mathbf{P}(X_i = 1) &= 7/20\\ \mathbf{P}(X_i = 0) &= 5/20\\ \mathbf{P}(X_i = -1) &= 7/20.\end{aligned}$$

Define $X_i' = \min(X_i, 1)$ for all i, and consider the *simple* random walk $S_n' = u + \sum_{i=1}^{n} X_i'$. Clearly $S_n \geq S_n'$ always, for all n, and so

$$\begin{aligned}\mathbf{P}(S_n \text{ ever hits } 0) &\leq \mathbf{P}(S_n' \text{ ever hits } 0)\\ &= \min\left(1, \left(\frac{7/20}{8/20}\right)^5\right).\end{aligned}$$

(b) i. A reasonable estimate of π is $\hat{\pi} = 29/100 = 0.29$.

ii. Using the binomial distribution with parameters 100 and π (estimated by $\hat{\pi}$—see the lecture notes), we obtain that the approximate standard deviation (standard error) of the above estimate is

$$\frac{1}{100}\sqrt{\frac{29(100-29)}{100}} = 0.045.$$

iii. Let n be the number of simulations. We require

$$\frac{1}{n}\sqrt{n\pi(1-\pi)} \leq 0.01,$$

where we estimate π by $\hat{\pi} = 0.29$. Rearranging, we have

$$\frac{\hat{\pi}(1-\hat{\pi})}{0.01^2} \leq n,$$

i.e.

$$n \geq 100^2 * 0.29(1-0.29) = 2,059.$$

4. (a) Note that if $0 < a < b$, then $T \geq 1$ since we need to go at least one step. It follows that $e_{b,a} = \mathbf{E}(T) \geq 1$ and so

$$T = 1 + T'$$

where T' is the remaining time after the first step to reach 0 or b, i.e.

$$T' = \min_{n \geq 0}\{n : S_{n+1} = 0 \text{ or } S_{n+1} = b\}.$$

Recall $\mathbb{I}(A)$ is the indicator random variable taking the value 1 if the event A occurs, and 0 otherwise, and that the key property of $\mathbb{I}(A)$ is that for any event A, we have

$$\mathbf{E}(\mathbb{I}(A)) = \mathbf{P}(A).$$

Arguing as in the lecture notes, we have, for $0 < a < b$,

$$\begin{aligned}
e_{b,a} &= \mathbf{E}(T \mid S_0 = a) = \mathbf{E}(1 + T' \mid S_0 = a) \\
&= 1 + \mathbf{E}(\mathbb{I}(S_1 = a+1)T' \mid S_0 = a) + \mathbf{E}(\mathbb{I}(S_1 = a-1)T' \mid S_0 = a) \\
&= 1 + \mathbf{E}(T' \mid S_0 = a, S_1 = a+1)\mathbf{E}(\mathbb{I}(S_1 = a+1) \mid S_0 = a) \\
&\quad + \mathbf{E}(T' \mid S_0 = a, S_1 = a-1)\mathbf{E}(\mathbb{I}(S_1 = a-1) \mid S_0 = a) \\
&= 1 + \mathbf{E}(T' \mid S_1 = a+1) * \frac{1}{2} + \mathbf{E}(T' \mid S_1 = a-1) * \frac{1}{2} \\
&= 1 + \frac{1}{2}\mathbf{E}(T \mid S_0 = a+1) + \frac{1}{2}\mathbf{E}(T \mid S_0 = a-1) \\
&= 1 + \frac{1}{2}e_{b,a+1} + \frac{1}{2}e_{b,a-1}.
\end{aligned}$$

(b) It is clear that the proposed solution satisfies the boundary conditions

$$e_{b,0} = e_{b,b} = 0.$$

Now let us verify that the above recurrence relation is satisfied by the given solution:

$$\begin{aligned}
1 + \frac{1}{2}e_{b,a+1} + \frac{1}{2}e_{b,a-1} &= 1 + \frac{1}{2}(a+1)(b-a-1) + \frac{1}{2}(a-1)(b-a+1) \\
&= a(b-a) = e_{b,a}.
\end{aligned}$$

(c) For fixed $a > 0$, the expected time to ruin is given by

$$\lim_{b \to \infty} e_{b,a} = \lim_{b \to \infty}(a(b-a)) = \infty.$$